

A Generalized Frey Curve with Perfect Square Discriminant: Insights into Beal-like Diophantine Equations

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Abstract

We introduce a generalized Frey curve associated with Beal-like equations of the form $a^n + b^m = c^k$ where $n, m, k > 2$ and $\gcd(a, b, c) = 1$. The curve is defined as $E_{n,m} : y^2 = x(x - a^n)(x + b^m)$. Our main contribution is the discovery that, whenever $a^n + b^m = c^k$, the discriminant of $E_{n,m}$ is a perfect square: $\Delta = (a^n b^m c^k)^2$.

This square-discriminant property establishes a direct algebraic link between Diophantine solutions and the invariants of the associated curve. We provide a complete algebraic proof of the property and analyze its implications for the conductor and potential modularity constraints. Special cases where the curve remains elliptic ($n = m = 2$ or $n = 2, m > 2$) are amenable to classical modularity techniques, while the general hyperelliptic case ($n, m > 2$) presents significant theoretical challenges.

Computational verification in limited ranges supports the absence of small counterexamples to the Beal Conjecture. The approach offers a unified framework for studying generalized Fermat-type equations and provides novel constraints that may contribute to future progress.

Keywords: Beal Conjecture, Frey curves, discriminant, elliptic curves, hyperelliptic curves, Diophantine equations, modularity

1. Introduction

The Beal Conjecture, formulated by Andrew Beal in 1993 while investigating generalizations of Fermat's Last Theorem, asserts that if $A^x + B^y = C^z$ where A, B, C, x, y, z are positive integers with $x, y, z > 2$, then A, B, C must share a common prime factor, i.e., $\gcd(A, B, C) > 1$ [1, 2].

The classical Frey curve approach, introduced by Gerhard Frey [3] and pivotal in Andrew Wiles' proof of Fermat's Last Theorem [4], associates to solutions of $A^p + B^p = C^p$ the elliptic curve $y^2 = x(x - A^p)(x + B^p)$. This curve exhibits special arithmetic properties that lead to contradictions with modularity theorems.

Inspired by this framework, we generalize the construction to equations with possibly distinct exponents: $a^n + b^m = c^k$. The resulting curve $E_{n,m} : y^2 = x(x - a^n)(x + b^m)$ unifies Fermat and Beal cases. When $n = m = k$, it reduces to the classical Frey curve.

Our key discovery is that whenever $a^n + b^m = c^k$, the discriminant Δ of $E_{n,m}$ is a perfect square: $\Delta = (a^n b^m c^k)^2$. This property provides a novel algebraic constraint linking Diophantine solutions to curve invariants.

We prove the square-discriminant property, analyze conductor properties under the coprimality condition $\gcd(a, b, c) = 1$, and discuss computational verification. While the hyperelliptic nature of $E_{n,m}$ for $n, m > 2$ limits direct application of elliptic curve modularity, special cases where the curve remains elliptic offer promising avenues for further investigation.

2. The Generalized Frey Curve

Definition 2.1. For positive integers a, n, b, m , the generalized Frey curve is

$$E_{n,m} : y^2 = x(x - a^n)(x + b^m).$$

Expanding the right-hand side yields the Weierstrass form

$$y^2 = x^3 + (b^m - a^n)x^2 - (a^n b^m)x.$$

The coefficients are

- $A = b^m - a^n$,
- $B = -a^n b^m$,
- $C = 0$.

When $n = m = k = p$, $E_{n,m}$ recovers the classical Frey curve associated to Fermat equations.

3. The Square-Discriminant Property

Theorem 3.1 (Square-Discriminant Property). Suppose $a^n + b^m = c^k$ for some positive integer c . Then the discriminant of $E_{n,m}$ is

$$\Delta = (a^n b^m c^k)^2.$$

Proof. The discriminant of a cubic $y^2 = x^3 + Ax^2 + Bx + C$ is

$$\Delta = A^2 B^2 - 4B^3 - 4A^3 C - 27C^2 + 18ABC.$$

Substituting $C = 0$ simplifies to

$$\Delta = A^2 B^2 - 4B^3 = B^2(A^2 - 4B).$$

Now substitute $A = b^m - a^n$ and $B = -a^n b^m$:

$$\Delta = (-a^n b^m)^2 \left[(b^m - a^n)^2 - 4(-a^n b^m) \right] = (a^n b^m)^2 \left[(b^m - a^n)^2 + 4a^n b^m \right].$$

Expanding the bracket:

$$(b^m - a^n)^2 + 4a^n b^m = b^{2m} - 2a^n b^m + a^{2n} + 4a^n b^m = a^{2n} + 2a^n b^m + b^{2m} = (a^n + b^m)^2$$

Thus,

$$\Delta = (a^n b^m)^2 (a^n + b^m)^2 = (a^n b^m c^k)^2,$$

since $a^n + b^m = c^k$. This completes the proof.

The perfect square discriminant is a striking algebraic feature that ties Diophantine solutions directly to the arithmetic invariants of the curve.

4. Conductor Analysis and Arithmetic Constraints

The conductor N of $E_{n,m}$ is determined by the primes of bad reduction, which divide the discriminant Δ . When $a^n + b^m = c^k$, the primes dividing Δ are precisely those dividing abc .

Under the coprimality condition $\gcd(a, b, c) = 1$, the primes dividing a , b , and c are distinct. This implies that the conductor N is square-free (or has low powers) and is a product of distinct primes.

In elliptic cases ($n = m = 2$ or $n = 2, m > 2$), the conductor may be amenable to level-lowering arguments similar to those in Wiles' proof. In the general hyperelliptic case, the conductor analysis suggests strong constraints on potential modular forms, though full modularity results remain unavailable.

5. Computational Verification

We implemented a search for solutions to $a^n + b^m = c^k$ with $\gcd(a, b, c) = 1$, bases up to 15, and exponents from 3 to 4. No counterexamples were found, consistent with the Beal Conjecture.

For the square-discriminant property, we verified explicit examples:

- $a = 2, n = 3, b = 1, m = 3, c = 3, k = 2: 8 + 1 = 9, \Delta = 5184 = 72^2$.
- Additional checks confirm the property holds algebraically and numerically.

The absence of small solutions and the consistent verification of the discriminant formula support the theoretical framework.

6. Relation to Existing Approaches

The square-discriminant property offers an algebraic counterpart to geometric constraints, such as those arising from Fermat angles. It also connects to the ABC conjecture through radical bounds on the conductor. In elliptic special cases, the approach aligns with classical Frey curve techniques [3, 4].

Work on multi-Frey curves and generalized Fermat equations [5, 6] provides related frameworks, though our square-discriminant property appears novel in this context.

7. Conclusion and Future Directions

We have introduced a generalized Frey curve with a remarkable square-discriminant property that links Diophantine solutions of Beal-like equations to algebraic curve invariants. While the property provides novel constraints, the hyperelliptic nature of the general case presents significant challenges for modularity theory.

Future work should focus on:

- Extending modularity results to hyperelliptic Jacobians,
- Developing p-adic Hodge theory applications,
- Exploring connections to Diophantine approximation and ABC-type bounds,
- Investigating elliptic special cases in greater depth.

The square-discriminant property itself is a mathematically noteworthy discovery that merits further study in the context of generalized Fermat equations.

References

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(Note: This bibliography includes foundational and related works. Additional references to recent preprints on Beal attempts are available upon request, though the conjecture remains open.)